

M21 2026 July 2026 Response Time

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Load libraries

Set emmeans options

```
emm_options(  
  lmer.df = "kenward-roger",  
  pbkrtest.limit = 10000,  
  disable.pbkrtest = FALSE  
)
```

Set ggplot parameters

```
# Color palette  
my_palette_1 <- c(  
  "#E31A1C", # red  
  "#1F78B4" # blue  
)  
  
my_palette_2 <- c(  
  "Simple" = "#0072B2",  
  "Complex" = "#E69F00"  
)  
  
my_linetypes_1 <- c(  
  "Simple" = "dashed",  
  "Complex" = "solid"  
)  
  
# Theme  
my_theme <- theme_classic(base_size = 10) +  
  theme(  
    legend.position = "bottom",  
    legend.direction = "horizontal",  
    legend.box = "horizontal"  
  )  
  
# Common guides  
my_guides <- guides(  
  colour = guide_legend(nrow = 1),  
  fill = guide_legend(nrow = 1)  
)
```

Define 2-way interaction table function

1 Load Files

```
# Load data: rt words  
rt_w <- read_csv("rt_words_frq.csv") |>
```

```

filter(correct == 1) |>
mutate(
  SubjID = factor(SubjID),
  Item = factor(target),
  logRT = log(response_time),
  OS = as.numeric(Dim.2),
  LP = as.numeric(Dim.1),
  zLogBF = as.numeric(scale(LogBF)),
  zLogFS = as.numeric(scale(LogFS))
)

# Load data: rt nonwords
rt_nw <- read_csv("rt_nwords_frq.csv") |>
filter(correct == 1) |>
mutate(
  SubjID = factor(SubjID),
  ItemID = factor(ItemID),
  complexity = factor(complexity,
    levels = c("SIMP", "COMP"),
    labels = c("Simple", "Complex")),
  logRT = log(response_time),
  OS = as.numeric(Dim.2),
  LP = as.numeric(Dim.1),
  zLogBF = as.numeric(scale(LogBF)),
  zLogFS = as.numeric(scale(LogFS))
)

# Load data: erp_fs words
erp_fs_w <- read_csv("erp_fs_dmg_words.csv") |>
mutate(
  SubjID = factor(SubjID),
  chlabel = factor(chlabel),
  family_size = factor(family_size,
    levels = c("Small", "Large")),
  time_window = factor(time_window),
  OS = as.numeric(Dim.2),
  LP = as.numeric(Dim.1)
)

# Load data: erp_fs nonwords
erp_fs_nw <- read_csv("erp_fs_dmg_nonwords.csv") |>
mutate(
  SubjID = factor(SubjID),
  chlabel = factor(chlabel),
  family_size = factor(family_size,
    levels = c("Small", "Large")),
  complexity = factor(complexity,
    labels = c("Simple", "Complex")),
  time_window = factor(time_window),
  OS = as.numeric(Dim.2),
  LP = as.numeric(Dim.1)
)

# Load data: erp_bf words
erp_bf_w <- read_csv("erp_bf_dmg_words.csv") |>
mutate(
  SubjID = factor(SubjID),
  chlabel = factor(chlabel),
  base_frequency = factor(base_frequency,
    levels = c("Low", "High")),
  time_window = factor(time_window),
  OS = as.numeric(Dim.2),
  LP = as.numeric(Dim.1)
)

# Load data: erp_bf nonwords
erp_bf_nw <- read_csv("erp_bf_dmg_nonwords.csv") |>
mutate(
  SubjID = factor(SubjID),
  chlabel = factor(chlabel),
  base_frequency = factor(base_frequency,
    levels = c("Low", "High")),
  complexity = factor(complexity,
    labels = c("Simple", "Complex")),
  time_window = factor(time_window),
  OS = as.numeric(Dim.2),
  LP = as.numeric(Dim.1)
)

```

```
range(erp_fs_nw$LP)

|| [1] -2.674523  3.111265
range(erp_fs_nw$OS)

|| [1] -1.975820  1.926921
```

2 RESPONSE TIME WORD DATA

2.1 Model

```
library(lmerTest)

m_rt_words <- lmer(
  logRT ~ zLogBF + zLogFS + OS + LP +
    (1 | SubjID) +
    (1 | Item),
  data = rt_w,
  control = lmerControl(optimizer = "bobyqa")
)
summary(m_rt_words)

|| Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
|| Formula: logRT ~ zLogBF + zLogFS + OS + LP + (1 | SubjID) + (1 | Item)
|| Data: rt_w
|| Control: lmerControl(optimizer = "bobyqa")
||
|| REML criterion at convergence: -3536.3
||
|| Scaled residuals:
||      Min       1Q   Median       3Q      Max
|| -9.8884 -0.6626 -0.0643  0.5951  5.1184
||
|| Random effects:
|| Groups   Name                Variance Std.Dev.
|| Item     (Intercept)  0.001807  0.04251
|| SubjID    (Intercept)  0.017718  0.13311
|| Residual                    0.029621  0.17211
|| Number of obs: 5847, groups: Item, 100; SubjID, 67
||
|| Fixed effects:
||              Estimate Std. Error    df t value Pr(>|t|)
|| (Intercept)   6.390069   0.016965  72.434536  376.655 < 2e-16 ***
|| zLogBF        -0.017184   0.004696  100.974079  -3.659 0.000404 ***
|| zLogFS        -0.012858   0.004865   93.591020  -2.643 0.009629 **
|| OS            -0.034712   0.018734   63.842814  -1.853 0.068518 .
|| LP             0.011591   0.012386   63.973484   0.936 0.352893
|| ---
|| Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
||
|| Correlation of Fixed Effects:
||      (Intr) zLogBF zLogFS OS
|| zLogBF  0.009
|| zLogFS  0.002 -0.105
|| OS      0.012 -0.001  0.001
|| LP      0.008  0.000  0.001  0.008

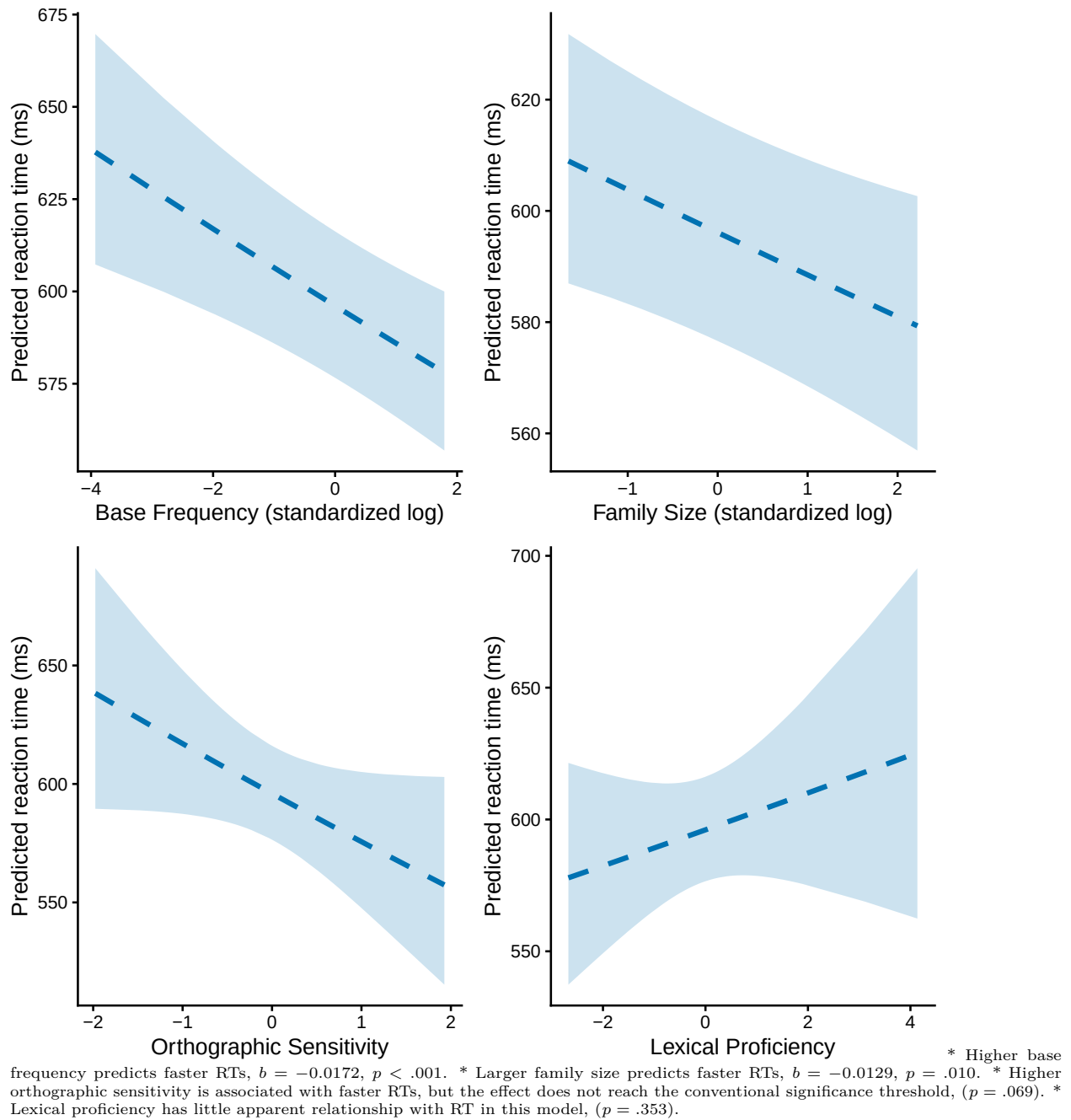
# Effect sizes for individual terms
eta_squared(m_rt_words, partial = TRUE)

|| # Effect Size for ANOVA (Type III)
||
|| Parameter | Eta2 (partial) |      95% CI
|| -----|-----|-----|
|| zLogBF    |      0.12 | [0.04, 1.00]
|| zLogFS    |      0.07 | [0.01, 1.00]
|| OS        |      0.05 | [0.00, 1.00]
|| LP        |      0.01 | [0.00, 1.00]
||
|| - One-sided CIs: upper bound fixed at [1.00].

# Model R²
r2_nakagawa(m_rt_words)

|| # R2 for Mixed Models
||
|| Conditional R2: 0.417
|| Marginal R2: 0.033
```

Adjusted Effects on Reaction Time



3 RESPONSE TIME NONWORD DATA

3.1 Model

```
contrasts(rt_nw$complexity) <- c(-0.5, 0.5)

m_rt_nw <- lmer(
  logRT ~ complexity + zLogBF + zLogFS + OS + LP +
    (1 + complexity | SubjID) +
    (1 | ItemID),
  data = rt_nw,
  control = lmerControl(optimizer = "bobyqa")
)
```

```
m_rt_nw
```

```
|| Linear mixed model fit by REML ['lmerModLmerTest']
|| Formula: logRT ~ complexity + zLogBF + zLogFS + OS + LP + (1 + complexity | SubjID) + (1 | ItemID)
|| Data: rt_nw
|| REML criterion at convergence: -4165.782
|| Random effects:
|| Groups Name Std.Dev. Corr
|| ItemID (Intercept) 0.03662
|| SubjID (Intercept) 0.13143
|| complexity1 0.02365 -0.17
|| Residual 0.14641
|| Number of obs: 4607, groups: ItemID, 100; SubjID, 67
|| Fixed Effects:
|| (Intercept) complexity1 zLogBF zLogFS OS LP
|| 6.554486 0.048807 0.015469 -0.006359 -0.041063 0.003398
```

```
summary(m_rt_nw)
```

```
|| Linear mixed model fit by REML. t-tests use Satterthwaite's method ['lmerModLmerTest']
|| Formula: logRT ~ complexity + zLogBF + zLogFS + OS + LP + (1 + complexity | SubjID) + (1 | ItemID)
|| Data: rt_nw
|| Control: lmerControl(optimizer = "bobyqa")
||
|| REML criterion at convergence: -4165.8
||
|| Scaled residuals:
|| Min 1Q Median 3Q Max
|| -4.2605 -0.6625 -0.0227 0.6633 3.8775
||
|| Random effects:
|| Groups Name Variance Std.Dev. Corr
|| ItemID (Intercept) 0.0013408 0.03662
|| SubjID (Intercept) 0.0172747 0.13143
|| complexity1 0.0005594 0.02365 -0.17
|| Residual 0.0214360 0.14641
|| Number of obs: 4607, groups: ItemID, 100; SubjID, 67
||
|| Fixed effects:
|| Estimate Std. Error df t value Pr(>|t|)
|| (Intercept) 6.554486 0.016639 68.832580 393.923 < 2e-16 ***
|| complexity1 0.048807 0.005347 62.675281 9.128 4.07e-13 ***
|| zLogBF 0.015469 0.005080 98.214341 3.045 0.00299 **
|| zLogFS -0.006359 0.005137 96.636227 -1.238 0.21880
|| OS -0.041063 0.018454 62.212180 -2.225 0.02970 *
|| LP 0.003398 0.012228 62.906744 0.278 0.78204
|| ---
|| Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
||
|| Correlation of Fixed Effects:
|| (Intr) cmplx1 zLogBF zLogFS OS
|| complexity1 -0.073
|| zLogBF -0.004 -0.005
|| zLogFS -0.004 0.011 -0.576
|| OS 0.011 -0.007 -0.001 -0.001
|| LP 0.003 -0.015 -0.002 0.000 0.009
```

```
# Effect sizes for individual terms
eta_squared(m_rt_nw, partial = TRUE)
```

```
|| # Effect Size for ANOVA (Type III)
||
|| Parameter | Eta2 (partial) | 95% CI
|| -----|-----|-----|
|| complexity | 0.57 | [0.44, 1.00]
|| zLogBF | 0.09 | [0.02, 1.00]
|| zLogFS | 0.02 | [0.00, 1.00]
|| OS | 0.07 | [0.00, 1.00]
|| LP | 1.23e-03 | [0.00, 1.00]
||
|| - One-sided CIs: upper bound fixed at [1.00].
```

```
# Model R2
r2_nakagawa(m_rt_nw)
```

```
|| # R2 for Mixed Models
||
|| Conditional R2: 0.494
|| Marginal R2: 0.050
```

```
# Means
# get means on log scale
```

```

emm_logrt <- emmeans(m_rt_nw, ~ complexity)
emm_summary <- summary(emm_logrt)

# back transform manually
emm_summary$mean_ms <- exp(emm_summary$emmean)
emm_summary$lower_ms <- exp(emm_summary$lower.CL)
emm_summary$upper_ms <- exp(emm_summary$upper.CL)
emm_summary

|| complexity emmean      SE    df lower.CL upper.CL mean_ms lower_ms upper_ms
|| Simple      6.529 0.0172 70.03   6.495   6.563   684.6   661.5   708.5
|| Complex     6.578 0.0168 70.15   6.544   6.611   718.9   695.2   743.4
||
|| Degrees-of-freedom method: kenward-roger
|| Confidence level used: 0.95
diff_ms <- exp(emm_summary$emmean[emm_summary$complexity == "Complex"]) -
            exp(emm_summary$emmean[emm_summary$complexity == "Simple"])
diff_ms

|| [1] 34.24345

```

For this model, use one adjusted-means plot for the categorical predictor `complexity` and four adjusted-effect plots for the continuous predictors. Because there are no interactions, the continuous-predictor slopes do not vary by complexity on the model's log-RT scale.

Then combine them:

```

# row1 <- plot_grid( BF_plot, FS_plot, OS_plot, LP_plot, ncol = 2)

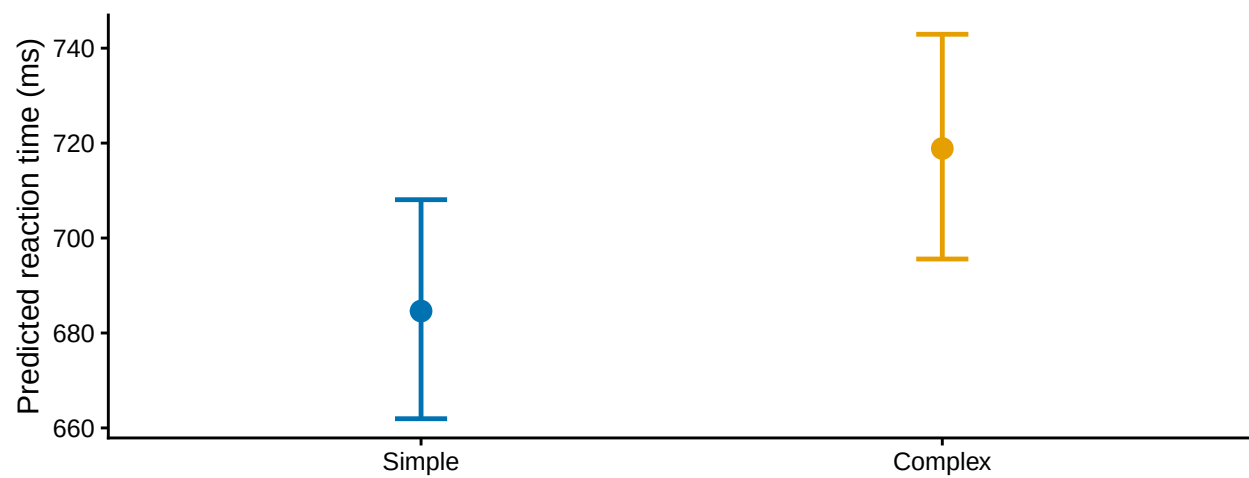
row1 <- plot_grid(
  BF_plot,
  FS_plot,
  ncol = 2,
  align = "hv"
)

row2 <- plot_grid(
  OS_plot,
  LP_plot,
  ncol = 2,
  align = "hv"
)

plot_grid(
  ggdraw() + draw_label("Nonword RT Data", fontface = "bold"),
  complexity_plot,
  row1,
  row2,
  ncol = 1,
  rel_heights = c(0.10, 1, 1, 1)
)

```

Nonword RT Data



Complexity

